Math 541B HW 3

Exercise 1. Let $X_1, \ldots, X_n$ be i.i.d. Gaussian random variables with unknown mean and unknown variance.

- Find a real-valued pivotal quantity for $X = (X_1, \ldots, X_n)$.
- Using the pivotal quantity, construct a $1 - \alpha$ confidence interval for the mean μ, for any $0 < \alpha < 1$.

Solution: 1) Let $X_1, X_2, \ldots, X_n$ be i.i.d. p-dimensional random vectors drawn from a multivariate normal distribution with an unknown mean vector $\mu \in \mathbb{R}^p$ and an unknown, positive-definite covariance matrix Σ.

We define the sample mean vector $\bar{X}$ and the unbiased sample covariance matrix S as:

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \quad \text{and} \quad S = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})(X_i - \bar{X})^T$$

A pivotal quantity is a function of the sample data and the unknown parameters whose probability distribution does not depend on any unknown parameters. For a multivariate vector parameter, the standard real-valued pivotal quantity is Hotelling's T^2 statistic.

From the properties of multivariate normal samples (assuming $n > p$):

- The sample mean vector follows a normal distribution: $\bar{X} \sim N_p\left(\mu, \frac{1}{n}\mathbf{\Sigma}\right)$, which implies:
$$\sqrt{n}(\bar{X} - \mu) \sim N_p(0, \mathbf{\Sigma})$$
- The scaled sample covariance matrix follows a Wishart distribution: $(n-1)S \sim W_p(n-1, \mathbf{\Sigma})$.
- $\bar{X}$ and S are statistically independent.

By scaling the squared Mahalanobis distance between the sample mean and the true population mean vector by the sample covariance matrix, the unknown population covariance matrix $\mathbf{\Sigma}$ cancels out. The pivotal quantity is defined as:

$$T^2 = n(\bar{X} - \mu)^T S^{-1}(\bar{X} - \mu)$$

The distribution of this statistic is uniquely determined. Specifically, a monotonic transformation of T^2 exactly follows an F-distribution:

$$\frac{n-p}{p(n-1)}T^2 \sim F_{p,\, n-p}$$

Since the distribution of T^2 depends only on the known constants n and p (and not on μ or $\mathbf{\Sigma}$), T^2 serves as our real-valued pivotal quantity.

2) Because μ is a multi-dimensional vector, its confidence structure forms a geometric hyper-ellipsoid in $\mathbb{R}^p$ space rather than a simple line interval.

Let $F_{\alpha,\,p,\,n-p}$ denote the upper α critical value of the F-distribution with p and $n-p$ degrees of freedom, such that $P(F > F_{\alpha,\,p,\,n-p}) = \alpha$. Using our pivotal distribution statement, we can write:

$$P\left(\frac{n-p}{p(n-1)} n(\bar{X}-\mu)^T S^{-1}(\bar{X}-\mu) \leq F_{\alpha,\,p,\,n-p}\right) = 1-\alpha$$

Isolating the quadratic form yields the boundary criteria for the parameter space. The $1-\alpha$ confidence region for the mean vector μ consists of all vectors μ satisfying the inequality:

$$\left\{\mu \in \mathbb{R}^p \;\middle|\; (\bar{X}-\mu)^T S^{-1}(\bar{X}-\mu) \leq \frac{p(n-1)}{n(n-p)} F_{\alpha,\,p,\,n-p}\right\}$$

Exercise 2. Let $X_1, \ldots, X_n$ be a real-valued random sample of size n from a family of distributions $\{f_\theta : \theta \in \Theta\}$. Suppose $\Theta = \mathbb{R}$. Fix $\theta \in \mathbb{R}$. Denote $X := (X_1, \ldots, X_n)$. Consider a set of nonrandomized hypothesis tests with rejection regions $C_\alpha \subseteq \mathbb{R}^n$ for all $\alpha \in [0,1]$. Suppose these rejection regions are nested in the sense that $C_\alpha \subseteq C_{\alpha'}$ for all $0 \leq \alpha < \alpha' \leq 1$. As usual, denote $\Theta = \Theta_0 \cup \Theta_1$ with $\Theta_0 \cap \Theta_1 = \varnothing$. Define also the p-valued

$$p(x) := \inf\{\alpha \in [0,1] : x \in C_\alpha\}, \qquad \forall\ x \in \mathbb{R}^n.$$

- Suppose $\sup_{\theta \in \Theta_0} \mathbb{P}_\theta(X \in C_\alpha) \leq \alpha$ for all $0 \leq \alpha \leq 1$. Show that the p-valued satisfies

$$\mathbb{P}_\theta(p(X) \leq c) \leq c, \qquad \forall\ 0 \leq c \leq 1,\ \forall\ \theta \in \Theta_0.$$

- Suppose $\mathbb{P}_\theta(X \in C_\alpha) = \alpha$ for all $0 \leq \alpha \leq 1$. Show that the p-valued satisfies

$$\mathbb{P}_\theta(p(X) \leq c) = c, \qquad \forall\ 0 \leq c \leq 1,\ \forall\ \theta \in \Theta_0.$$

  That is, $p(X)$ is uniformly distributed in $[0,1]$.

Solution: 1) To show that $P_\theta(p(X) \leq c) \leq c$ for any $c \in [0,1]$ and $\theta \in \Theta_0$, we establish a set equivalence relationship between the event $\{p(X) \leq c\}$ and the nested rejection regions.

By definition, the p-value is $p(x) = \inf\{\alpha \in [0,1] : x \in C_\alpha\}$. If $p(x) \leq c$, then for any $\epsilon > 0$, there exists an $\alpha \leq c + \epsilon$ such that $x \in C_\alpha$. Since the rejection regions are nested ($C_\alpha \subseteq C_{\alpha'}$ for $\alpha < \alpha'$), it follows that $x \in C_{c+\epsilon}$.

If $p(x) \leq c$, then $x \in C_{c+\epsilon}$ for all $\epsilon > 0$. Therefore, we have the following set containment:

$$\{x \in \mathbb{R}^n : p(x) \leq c\} \subseteq \bigcap_{\epsilon > 0} C_{c+\epsilon}$$

For any fixed $\epsilon > 0$, the containment $\{p(X) \leq c\} \subseteq C_{c+\epsilon}$ holds. Taking the probability under any $\theta \in \Theta_0$:

$$P_\theta(p(X) \leq c) \leq P_\theta(X \in C_{c+\epsilon})$$

We are given that $\sup_{\theta \in \Theta_0} P_\theta(X \in C_\alpha) \leq \alpha$ for all $\alpha \in [0,1]$. Thus, for any specific $\theta \in \Theta_0$:

$$P_\theta(X \in C_{c+\epsilon}) \leq c + \epsilon$$

Combining this with the previous inequality gives:

$$P_\theta(p(X) \leq c) \leq c + \epsilon$$

Since this inequality holds for all $\epsilon > 0$, we can let ϵ approach 0 from above, which yields:

$$P_\theta(p(X) \leq c) \leq c, \quad \forall 0 \leq c \leq 1, \forall \theta \in \Theta_0$$

2) To show that $p(X) \sim \text{Uniform}[0,1]$ under the condition $P_\theta(X \in C_\alpha) = \alpha$, we sandwich the probability of interest using the properties of the infimum and the nested sets.

Since $P_\theta(X \in C_\alpha) = \alpha \leq \alpha$, the condition $\sup_{\theta \in \Theta_0} P_\theta(X \in C_\alpha) \leq \alpha$ automatically holds. Therefore, from Part 1, we already possess the upper bound:

$$P_\theta(p(X) \leq c) \leq c$$

If $x \in C_c$, then by definition, $c \in \{\alpha \in [0,1] : x \in C_\alpha\}$. Because the p-value $p(x)$ is the infimum (greatest lower bound) of this set, $p(x)$ must be less than or equal to c. This gives us the reverse set containment:

$$C_c \subseteq \{x \in \mathbb{R}^n : p(x) \leq c\}$$

By the monotonicity property of probability measures:

$$P_\theta(X \in C_c) \leq P_\theta(p(X) \leq c)$$

We are given that $P_\theta(X \in C_\alpha) = \alpha$ for all $\alpha \in [0,1]$ and $\theta \in \Theta_0$. Substituting $\alpha = c$:

$$c \leq P_\theta(p(X) \leq c)$$

Combining our upper bound from Step 1 and our lower bound from Step 4:

$$c \leq P_\theta(p(X) \leq c) \leq c$$

Thus, it must hold exactly that:

$$P_\theta(p(X) \leq c) = c, \quad \forall 0 \leq c \leq 1, \forall \theta \in \Theta_0$$

Since its Cumulative Distribution Function (CDF) is $F(c) = c$ on the interval $[0,1]$, $p(X)$ is uniformly distributed on $[0,1]$. ■

Exercise 3. Let $X_1, \ldots, X_n$ be a random sample from an exponential distribution with unknown location parameter $\theta > 0$, i.e. X_1 has density

$$g(x) := \mathbf{1}_{x \geq \theta} e^{-(x-\theta)}, \qquad \forall\ x \in \mathbb{R}.$$

Fix $\theta_0 \in \mathbb{R}$. Suppose we want to test that hypothesis H_0 that $\theta \leq \theta_0$ versus the alternative H_1 that $\theta > \theta_0$. That is, $\Theta = \mathbb{R}$, $\Theta_0 = \{\theta \in \mathbb{R} : \theta \leq \theta_0\}$ and $\Theta_0^c = \Theta_1 = \{\theta \in \mathbb{R} : \theta > \theta_0\}$.

- Explicitly describe the rejection region of the generalized likelihood ratio test for this hypothesis. (Hint: it might be easier to describe the region using $x_{(1)} = \min(x_1, \ldots, x_n)$.)
- Prove that $X_{(1)} := \min(X_1, \ldots, X_n)$ is a sufficient statistic for θ.
- (Optional) If H_0 is true, then does

$$2 \log \frac{\sup_{\theta \in \Theta} f_\theta(X_1, \ldots, X_n)}{\sup_{\theta \in \Theta_0} f_\theta(X_1, \ldots, X_n)}$$

converge in distribution to a chi-squared distribution as $n \to \infty$?

Solution: The joint density (likelihood function) for a random sample $X = (X_1, \ldots, X_n)$ is given by:

$$f_\theta(x) = \prod_{i=1}^{n} \mathbf{1}_{\{x_i \geq \theta\}} e^{-(x_i - \theta)} = \mathbf{1}_{\{x_{(1)} \geq \theta\}} e^{-\sum_{i=1}^{n} x_i + n\theta}$$

where $x_{(1)} = \min(x_1, \ldots, x_n)$.

To find the generalized likelihood ratio $\Lambda(x) = \frac{\sup_{\theta \in \Theta} f_\theta(x)}{\sup_{\theta \in \Theta_0} f_\theta(x)}$, we evaluate the numerator and denominator separately.

To maximize $f_\theta(x)$, we maximize $e^{n\theta}$ subject to the indicator constraint $\theta \leq x_{(1)}$. Since $e^{n\theta}$ is strictly increasing in θ, the global Maximum Likelihood Estimator (MLE) is:

$$\hat{\theta} = x_{(1)}$$

Thus, the unconstrained supremum is:

$$\sup_{\theta \in \Theta} f_\theta(x) = f_{\hat{\theta}}(x) = e^{-\sum_{i=1}^{n} x_i + n x_{(1)}}$$

We maximize $f_\theta(x)$ subject to both $\theta \leq \theta_0$ and $\theta \leq x_{(1)}$, which requires $\theta \leq \min(\theta_0, x_{(1)})$. Due to the monotonicity of $e^{n\theta}$, the restricted MLE under H_0 is:

$$\hat{\theta}_0 = \min(\theta_0, x_{(1)})$$

Thus, the constrained supremum is:

$$\sup_{\theta \in \Theta_0} f_\theta(x) = f_{\hat{\theta}_0}(x) = e^{-\sum_{i=1}^{n} x_i + n \min(\theta_0, x_{(1)})}$$

Dividing the two suprema yields:

$$\Lambda(x) = \frac{e^{-\sum_{i=1}^{n} x_i + n x_{(1)}}}{e^{-\sum_{i=1}^{n} x_i + n \min(\theta_0, x_{(1)})}} = e^{n\left[x_{(1)} - \min(\theta_0, x_{(1)})\right]}$$

We evaluate this expression based on the value of $x_{(1)}$ relative to θ_0:

- If $x_{(1)} \leq \theta_0$, then $\min(\theta_0, x_{(1)}) = x_{(1)}$, making $\Lambda(x) = e^0 = 1$.

- If $x_{(1)} > \theta_0$, then $\min(\theta_0, x_{(1)}) = \theta_0$, making $\Lambda(x) = e^{n(x_{(1)}-\theta_0)}$.

The GLRT rejects H_0 if $\Lambda(x) > c$ for some critical value $c > 1$. This occurs only when $x_{(1)} > \theta_0$:

$$e^{n(x_{(1)}-\theta_0)} > c \implies x_{(1)} > \theta_0 + \frac{\log c}{n}$$

Hence, the explicit rejection region is:

$$C = \left\{x \in \mathbb{R}^n : x_{(1)} > k\right\} \quad \text{where } k = \theta_0 + \frac{\log c}{n} > \theta_0$$

We use the Fisher-Neyman Factorization Theorem, which states that $T(X)$ is sufficient for θ if the joint likelihood can be factored into a product $g(T(x), \theta) \cdot h(x)$.

From the likelihood function derived above, we can write:

$$f_\theta(x) = \mathbf{1}_{\{x_{(1)} \geq \theta\}} e^{n\theta} \cdot e^{-\sum_{i=1}^n x_i}$$

Let us define:

- $g(x_{(1)}, \theta) = \mathbf{1}_{\{x_{(1)} \geq \theta\}} e^{n\theta}$, which depends on x only through the statistic $x_{(1)}$.
- $h(x) = e^{-\sum_{i=1}^n x_i}$, which depends solely on the sample data x and is completely independent of θ.

Because the joint density factors cleanly into $g(x_{(1)}, \theta)h(x)$, $X_{(1)} = \min(X_1, \ldots, X_n)$ is a sufficient statistic for θ.

The test statistic of interest is $W_n = 2 \log \Lambda(X) = 2n \left[X_{(1)} - \min(\theta_0, X_{(1)})\right]$.

No, Wilks' Theorem fails here, and W_n does not automatically converge to a standard χ^2_1 distribution across all of H_0. Wilks' theorem requires the support of the distribution to be independent of the unknown parameter θ. Since the support is $X_i \geq \theta$, this is a non-regular estimation problem.

When $\theta = \theta_0$, the true parameter sits on the boundary. Because $X_i \geq \theta_0$ almost surely, it follows that $X_{(1)} \geq \theta_0$ with probability 1. Thus, $\min(\theta_0, X_{(1)}) = \theta_0$, simplifying our statistic to:

$$W_n = 2n(X_{(1)} - \theta_0)$$

The exact survival function of $X_{(1)}$ under θ_0 is:

$$P_{\theta_0}(X_{(1)} > x) = [P_{\theta_0}(X_1 > x)]^n = \left(e^{-(x-\theta_0)}\right)^n = e^{-n(x-\theta_0)} \quad \text{for } x \geq \theta_0$$

Now, we find the Cumulative Distribution Function (CDF) of W_n for any $w > 0$:

$$P_{\theta_0}(W_n \leq w) = P_{\theta_0}\left(2n(X_{(1)} - \theta_0) \leq w\right) = P_{\theta_0}\left(X_{(1)} - \theta_0 \leq \frac{w}{2n}\right)$$

$$= 1 - P_{\theta_0}\left(X_{(1)} > \theta_0 + \frac{w}{2n}\right) = 1 - e^{-n\left(\frac{w}{2n}\right)} = 1 - e^{-w/2}$$

The CDF $F(w) = 1 - e^{-w/2}$ is the exact distribution function of an Exponential distribution with mean 2, which is identical to a $\chi^2(2)$ distribution with 2 degrees of freedom. This holds exactly for any sample size n, meaning the asymptotic distribution is χ^2_2.

By the properties of the sample minimum, $X_{(1)} \xrightarrow{P} \theta$ as $n \to \infty$. Since $\theta < \theta_0$, the probability that $X_{(1)} < \theta_0$ approaches 1 as $n \to \infty$. When $X_{(1)} \leq \theta_0$, the statistic $W_n = 0$. Consequently:

$$W_n \xrightarrow{d} 0$$

The distribution collapses to a degenerate distribution concentrated entirely at 0.

So under H_0 , the statistic converges to a χ^2_2 distribution only at the boundary point $\theta = \theta_0$, for any other parameter value in the null hypothesis ($\theta < \theta_0$) it converges to a constant 0.

Exercise 4. Let $X_1, \ldots, X_n$ be a random sample from a Gaussian random variable with unknown mean $\mu \in \mathbb{R}$ and unknown variance $\sigma^2 > 0$.

Fix $\mu_0 \in \mathbb{R}$. Suppose we want to test that hypothesis H_0 that $\mu = \mu_0$ versus the alternative H_1 that $\mu \neq \mu_0$.

- Explicitly describe the rejection region of the generalized likelihood ratio test for this hypothesis.
- Give an explicit formula for the p-value of this hypothesis test. (Hint: If S^2 denotes the sample variance and $\overline{X}$ denotes the sample mean, you should then be able to use the statistic

$$\frac{(\overline{X} - \mu_0)^2}{S^2}.$$

  Since we have an explicit formula for Snedecor's distribution, you should then be able to write an explicit integral formula for the p-value of this test.)

Solution: Let $X_1, \ldots, X_n$ be a random sample from a Gaussian random variable with unknown mean $\mu \in \mathbb{R}$ and unknown variance $\sigma^2 > 0$. We wish to test the null hypothesis:

$$H_0 : \mu = \mu_0 \quad \text{versus} \quad H_1 : \mu \neq \mu_0$$

where $\mu_0 \in \mathbb{R}$ is fixed.

The Generalized Likelihood Ratio Test (GLRT) statistic Λ is defined as:

$$\Lambda(X_1, \ldots, X_n) = \frac{\sup_{\theta \in \Theta_0} L(\mu, \sigma^2)}{\sup_{\theta \in \Theta} L(\mu, \sigma^2)}$$

The likelihood function for the sample is:

$$L(\mu, \sigma^2) = \left(2\pi\sigma^2\right)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2\right)$$

Under the unrestricted parameter space Θ, the maximum likelihood estimators are:

$$\hat{\mu} = \bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \quad \text{and} \quad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

Substituting these into the likelihood yields the maximized unrestricted likelihood:

$$\sup_{\theta\in\Theta} L(\mu, \sigma^2) = \left(2\pi\hat{\sigma}^2\right)^{-n/2} \exp\left(-\frac{n}{2}\right)$$

Under the restricted parameter space Θ_0 (where $\mu = \mu_0$), the restricted MLE for the variance is:

$$\hat{\sigma}_0^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu_0)^2$$

Substituting this yields the maximized restricted likelihood:

$$\sup_{\theta\in\Theta_0} L(\mu, \sigma^2) = \left(2\pi\hat{\sigma}_0^2\right)^{-n/2} \exp\left(-\frac{n}{2}\right)$$

Taking the ratio, the profile likelihood ratio simplifies to:

$$\Lambda = \left(\frac{\hat{\sigma}^2}{\hat{\sigma}_0^2}\right)^{n/2} = \left(\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{\sum_{i=1}^{n}(X_i - \mu_0)^2}\right)^{n/2}$$

Using the algebraic identity $\sum_{i=1}^{n}(X_i - \mu_0)^2 = \sum_{i=1}^{n}(X_i - \bar{X})^2 + n(\bar{X} - \mu_0)^2$, we can rewrite Λ as:

$$\Lambda = \left(1 + \frac{n(\bar{X} - \mu_0)^2}{\sum_{i=1}^{n}(X_i - \bar{X})^2}\right)^{-n/2} = \left(1 + \frac{n(\bar{X} - \mu_0)^2}{(n-1)S^2}\right)^{-n/2}$$

where $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$ denotes the unbiased sample variance.

The GLRT rejects H_0 if $\Lambda \leq c$ for some constant $c \in (0, 1)$. Since Λ is a strictly decreasing function of the statistic $\frac{(\bar{X}-\mu_0)^2}{S^2}$, the rejection condition is equivalent to:

$$\frac{(\bar{X} - \mu_0)^2}{S^2} > c'$$

Under H_0, the statistic $F = \frac{n(\bar{X}-\mu_0)^2}{S^2}$ follows Snedecor's F-distribution with 1 and $n - 1$ degrees of freedom ($F \sim F(1, n-1)$). Therefore, at a significance level α, the explicit rejection region is:

$$\mathcal{R} = \left\{(X_1, \ldots, X_n) \in \mathbb{R}^n : \frac{(\bar{X} - \mu_0)^2}{S^2} > \frac{1}{n}F_{1-\alpha}(1, n-1)\right\}$$

where $F_{1-\alpha}(1, n-1)$ is the $(1-\alpha)$-quantile of Snedecor's $F(1, n-1)$ distribution.

Let $\bar{X}_{\text{obs}}$ and S^2_{obs} be the observed sample mean and sample variance evaluated from the data. The observed test statistic value is:

$$F_{\text{obs}} = \frac{n(\bar{X}_{\text{obs}} - \mu_0)^2}{S^2_{\text{obs}}}$$

The p-value is the probability under H_0 of observing an F statistic at least as extreme as F_{obs}:

$$p\text{-value} = \mathbb{P}_{H_0}(F \geq F_{\text{obs}}) = \int_{F_{\text{obs}}}^{\infty} f(k)\, dk$$

where $f(k)$ is the probability density function of Snedecor's $F(1, n-1)$ distribution. The explicit density is given by:

$$f(k) = \frac{1}{\sqrt{n-1}\ \mathrm{B}\left(\frac{1}{2}, \frac{n-1}{2}\right)} k^{-1/2} \left(1 + \frac{k}{n-1}\right)^{-n/2}, \quad k > 0$$

where $\mathrm{B}(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$ is the Beta function.

Substituting $f(k)$ yields the explicit integral formula for the p-value:

$$p\text{-value} = \frac{1}{\sqrt{n-1}\ \mathrm{B}\left(\frac{1}{2}, \frac{n-1}{2}\right)} \int_{\frac{n(\bar{X}_{\text{obs}} - \mu_0)^2}{S^2_{\text{obs}}}}^{\infty} k^{-1/2} \left(1 + \frac{k}{n-1}\right)^{-n/2} dk$$